Assume/Guarantee Contract of `AttiCal`

May 29, 2026

1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet : workmode, flgSP, royaw, piyaw, W, angle′, rate′, Gi, Gi′, clock, clock′

Inputs :

- `workmode` $\mapsto$ workmode : $\mathbb{N}$
- `flgSP` $\mapsto$ flgSP : $\mathbb{N}$
- `royaw` $\mapsto$ royaw : $\mathbb{R}$
- `piyaw` $\mapsto$ piyaw : $\mathbb{R}$
- `W` $\mapsto$ W : $\mathbb{N} \rightharpoonup \mathbb{R}$

Outputs :

- `angle` $\mapsto$ angle : $\mathbb{N} \rightharpoonup \mathbb{R}$
- `rate` $\mapsto$ rate : $\mathbb{N} \rightharpoonup \mathbb{R}$

In-Outputs :

- `Gi` $\mapsto$ Gi : $\mathbb{N} \rightharpoonup \mathbb{R}$

States : None

Parameters : None

Clocks : clock, clock′ : $\mathbb{R}_{\geq 0}$

2 AG Contract

Assumptions :

$$Assumptions \triangleq \mathsf{len}(\mathsf{Gi}) = 2 \ \wedge \ \mathsf{len}(\mathsf{W}) = 3$$

Guarantees :

$$\begin{aligned}
Guarantees &\triangleq Check_1 \ \wedge \ Check_2 \\
Check_1 &\triangleq \mathsf{len}(\mathsf{angle}') = \mathsf{len}(\mathsf{Gi}') = 2 \\
&\quad \wedge \mathsf{len}(\mathsf{rate}') = 3 \\
&\quad \wedge \mathsf{rate}' = \mathsf{W} \\
&\quad \wedge \mathsf{clock}' = \mathsf{clock} \\
Check_2 &\triangleq \mathsf{if}\ \mathsf{workmode} = \texttt{0x33}\ \mathsf{then} \\
&\quad \mathsf{if}\ \mathsf{flgSP} = \texttt{0xEB}\ \mathsf{then} \\
&\quad\quad \mathsf{angle}' = \mathsf{Gi}' = \{0 \mapsto \mathsf{royaw}, 1 \mapsto \mathsf{piyaw}\} \\
&\quad \mathsf{else} \\
&\quad\quad \mathsf{angle}' = \mathsf{Gi}' = \mathsf{Gi} \\
&\quad \mathsf{endif} \\
&\mathsf{else} \\
&\quad \mathsf{angle}' = \{0 \mapsto 0, 1 \mapsto 0\} \\
&\quad \wedge \mathsf{Gi}' = \mathsf{Gi} \\
&\mathsf{endif}
\end{aligned}$$